AI RESEARCHER TAKE-HOME · WRITE-UP

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Evaluating LLM summaries of financial news

Can an analyst read the summary instead of the article, and trust it? Two approaches, a five-layer evaluation, 49 articles, 5 tickers.

FAITHFULNESS, 1–5
2.64 → **4.49**
Δ +1.85 [+1.56, +2.13]

UNSUPPORTED CLAIMS
3.87 → **0.64**
per summary

OVERSTATED TICKER LINK
44% → **0%**
share of summaries

PAIRWISE WINS
B 45 / A 3
of 49, order-swap stable

01 What the data demands

Short. Median 130 informative words: no chunking; the work is what to extract.

Often not about the ticker. 19 of 49 (39%) are wraps or roundups; 3 never mention it.

Thin. 11 (22%) carry the news only in the headline.

Numeric. Figures must be exact, so they get a deterministic check.

Both approaches use claude-sonnet-5 on the same input; A is a strong baseline that also knows the ticker.

02 Two approaches

	● A · Simple	◆ B · Structured
Output	Prose, ≤100 words	JSON: relevance, ≤50-word summary, 3 key facts, sentiment
Grounding	Implicit	Article-only, figures verbatim
Relevance	Ticker in prompt	4-level triage, says when peripheral

03 Evaluation framework: five layers, cheapest and most objective first

L1 · Deterministic
Every figure must match the article; reversed old/new pairs.

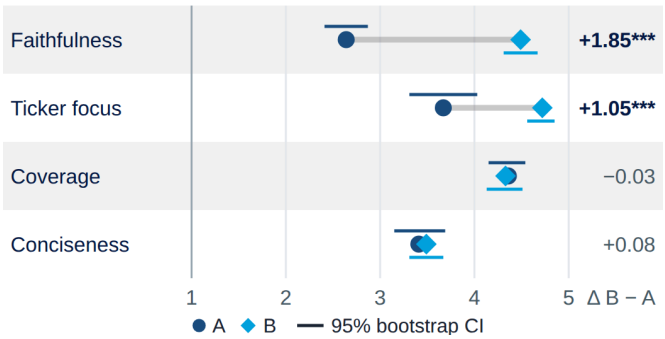
L2 · Human labels
Hand-labelled relevance for all 49 pairs.

L3 · Judge rubric
A stronger model lists unsupported claims, then scores 1–5.

L4 · Pairwise
Asked in both orders; flips don't count.

L5 · Test the judge
Injected errors, repeat-run noise, blinded review.

04 Results



Mean judge score, paired n = 39; *** Wilcoxon p < 0.001. Figures matching the article: A 95.9%, B 98.8%.

WHAT WORKED

Grounding removed A's main errors: speculation and outside facts.

Triage ended invented ticker links; labels match humans on 43/49.

No coverage cost at 37% fewer words (60 vs 96).

WHAT DIDN'T, OR ONLY PARTLY

Wraps over-rated: all 4 label errors are peripheral → secondary.

Judge alone isn't enough: it flags 58% of injected direction flips; the numeric check catches 92% of number swaps.

Run gaps (2 invalid JSON, 10 truncated judge replies) fixed via structured outputs.

05 Recommendations

1 **Adopt B's design:** grounded, triaged, schema-constrained; filter by relevance first.

2 **Make traceability the bar:** cite the source passage per claim.

3 **Route templated wires** (price-target notes) to rules.

4 **Monitor in layers:** numeric checks on 100%, validated judge on a sample.

LIMITATIONS · NEXT STEPS

- Small sample: 49 articles, 5 tickers, 2 months.
- One model family throughout → Multi-model, multi-prompt.
- One annotator → two analysts calibrate the judge.
- At scale: dedupe stories, cheap triage first.
- Keep a full audit trail and clear content

First full run; A's paired n = 39 (10 truncated judge replies). All LLM responses cached.

Notebook & code: